

SES598 Midterm Report

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1 Project Goal

The goal of this project is to create a multi-robot system that uses SLAM to understand individual position while building a shared map of an unknown environment. Robots use this map to identify unexplored frontiers and coordinate movement to maximize information gain while minimizing overlap. Overall, this is investigating the improved mapping efficiency of a multi-robot system.

2 Approach

To develop this project, the following approach was utilized. All in-progress items are in development for the final project submission.

1. Create working gazebo world: **Complete**
2. Spawn a single turtlebot3 waffle robot: **Complete**
3. Implement 2D SLAM on one robot: **Complete**
4. Spawn several robots: **Complete**
5. Merge the maps together of all the robots: **In-Progress**
6. Implement robot autonomous exploration: **In-Progress**
7. Unexplored Frontier Detection: **In-Progress**
8. Task allocation for mapping efficiency: **In-Progress**

The initial simulation setup will only utilize built in environments with obstacles provided by gazebo. The final simulation aims to utilize a Mars like simulated environment.

ROS2 has published extensive resources in relation to SLAM and has many pre-built packages

that can be utilized for this project. The core packages used are: `nav2_minimal_tb3_sim` [3], `slam_toolbox` [4], and `nav2_bringup` [5]. For package creation information, [6] was referenced in addition to other packages installed from ROS2.

Please note that AI resources were used in developing this code. Specifically, Claude AI was used to aid in compiling all relevant information and resources for ROS2, Nav2, and SLAM for implementation. Claude was also used to aid in resolving any error's encountered in code development.

3 Deliverables

Deliverables for Midterm Report:

1. Midterm Report: This will include the initial findings, resources, and preliminary results.
2. Github Repo of Initial Code Development: This will include the initial code used to spawn a single robots with the ability to create a 2D map of an area and the capability to spawn several robots.
3. Video Submission: Video of a single robot mapping an world in gazebo with obstacles.

Deliverables for Final Project:

1. Final Report: This will include the processes, metrics, final results, and snapshots of the implemented simulation for task allocation and improved frontier mapping efficiency of a multi-robot system.
2. Github Repo: This will include the final code, videos, and report.
3. Video Submission: Video of a multi-robot systems mapping a martian world in gazebo with rock obstacles.

4 Preliminary Results

Working Gazebo Obstacle Worlds:

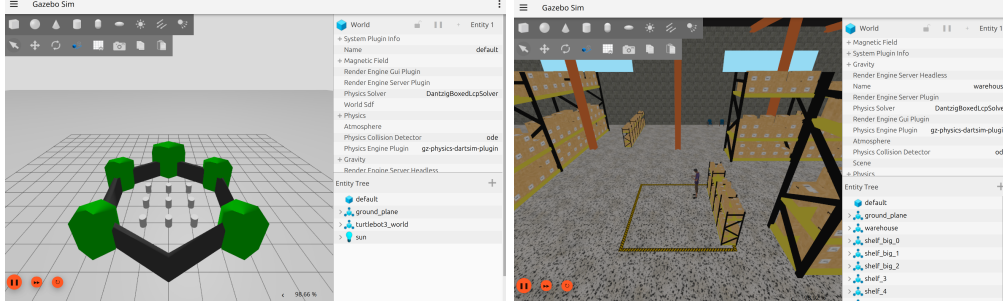


Figure 1: Provided Gazebo Obstacle Worlds

Spawning a Single Turtlebot3 Waffle and 2D SLAM Implementation:

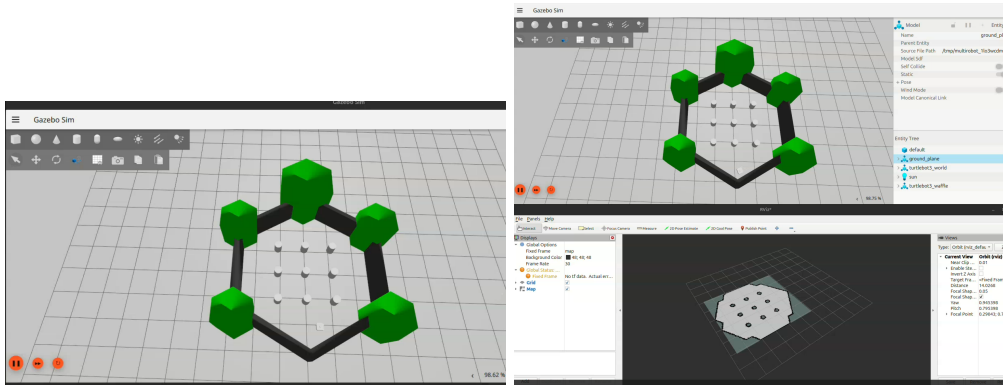


Figure 2: Single Robot Spawning and 2D SLAM

Multi-Robot Spawning:

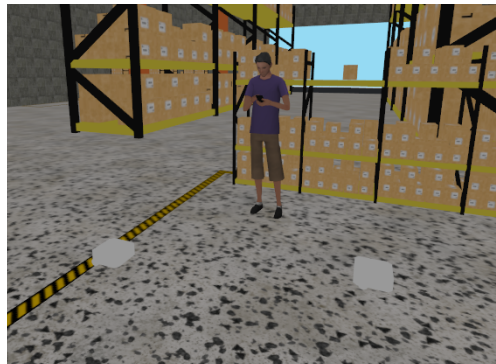


Figure 3: Multi-robot Spawning

Mars Surface Gazebo World: Please see reference [7] for more information and source Github.

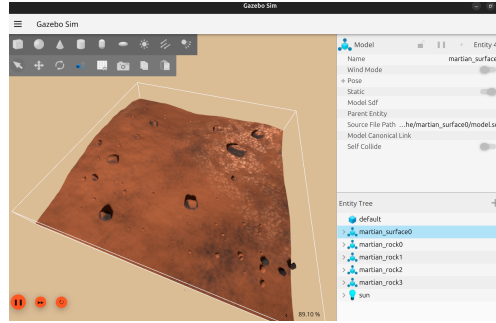


Figure 4: Simulated Martian World

5 References

- [1] Github Repository: <https://github.com/gciaglia/multirobotSLAM-ses598-finalproject>
- [2] Video Submission: <https://github.com/gciaglia/multirobotSLAM-ses598-finalproject/tree/main/MidtermSubmission/VideoSubmissions>
- [3] nav2_minimal_tb3_sim: https://github.com/ros-navigation/nav2_minimal_turtlebot_simulation/tree/main/nav2_minimal_tb3_sim/launch
- [4] slam_toolbox: https://github.com/SteveMacenski/slam_toolbox/tree/ros2
- [5] nav2_bringup: https://github.com/ros-navigation/navigation2/blob/humble/nav2_bringup/README.md
- [6] Package Creation Instructions: <https://docs.ros.org/en/jazzy/Tutorials/Beginner-Client-Libraries/Creating-Your-First-ROS2-Package.html>
- [7] Martian Environment: https://github.com/AndrejOrsula/space_robotics_gz_envs